LOCAL EXISTENCE THEORY OF CLASSICAL AND QUANTUM LANDAU EQUATIONS

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Abstract. The Landau equation is a kinetic equation which describes particles interacting through a Coulomb potential. We give an introduction to the homogeneous Landau equation and its local existence and regularity theory for bounded and L^p initial data. We will then discuss current work extending these results to the Landau–Bose–Einstein equation, which incorporates the quantum crowding effects of bosons.

Notes for a talk given at the UT Austin Junior Analysis seminar on 10/30.

1. Intro

Kinetic equations describe many particle systems in terms of a density function $f(x, v) \geq 0$ on phase space. Coarser than a particle level description, finer than a fluid level description. The Landau equation is a kinetic equation describing the evolution of a collisional plasma:

$$\partial_t f + v \cdot \nabla_x f = \nabla_v \cdot \int \frac{\Pi(v-w)}{|v-w|}[f(w)\nabla_v f(v) - f(v)\nabla_w f(w)]\, dw$$

where

$$\Pi(z) = I - \hat{z} \otimes \hat{z}.$$

It's derived (if you're a mathematician) by taking an appropriate limit of the Boltzmann equation for Coulomb (inverse square law) potentials. In this talk, I'm gonna introduce the basic properties and the local existence and regularity theory for the Landau equation in the homogeneous case. Then I'm going to discuss some work I've been doing to try to extend these results to the Landau–Bose–Einstein equation.

2. A Priori Estimates

The Landau equation conserves mass, momentum, and energy

$$\frac{d}{dt}\int \begin{pmatrix} 1 \\ v \\ |v|^2 \end{pmatrix} f\, dv = 0$$

The entropy

$$S[f] := \int f \log f\, dv$$

is also monotone decreasing, this is Boltzmann's H-theorem. A fifth quantity, the Fisher information

$$\int \frac{|\nabla f|^2}{f}\, dv$$

was shown to be monotone by Guillen and Silvestre (2023). We won't discuss the proof of this here, but we're going to go over local theory/continuation criteria that allow Fisher to imply global wellposedness. Equilibria are Maxwellians (Gaussians).

3. Short Time Classical L^∞ Theory

Theorem. *(Arsen'ev-Peskov 1975) If $f_0 \in L^\infty$, then the Landau equation exists and smooths up to a short time $T(\|f_0\|_\infty)$.*

The homogeneous Landau equation can be rewritten as a parabolic equation like so:

$$\partial_t f = \nabla \cdot (A[f]\nabla f - \nabla a[f] f)$$

where the coefficients are

$$A[f] = \frac{1}{|v|}(I - \hat{v} \otimes \hat{v}) * f, \nabla a[f] = \frac{v}{|v|^3} * f$$

$A[f]$ is degenerate elliptic with

$$A[f] \gtrsim \langle v \rangle^{-3} I.$$

Difficulty for existence theory is that these coefficients are not bounded given just the classical a priori estimates. The Hardy-Littlewood-Sobolev inequality and interpolation give us for $p > 3/2$

$$\|A[f]\|_\infty \lesssim \|f\|_1^{\frac{2p-3}{3p-3}} \|f\|_p^{\frac{p}{3p-3}}$$

and similar for $\nabla a[f]$. We control L^1 globally in time but not L^p, so we need to propagate the norm in time.

Write equation in nondivergence form

$$\partial_t f = \sum_{i,j} A[f]_{ij} \partial_{ij} f + f^2.$$

Leads to formal estimate

$$\frac{d}{dt}\|f\|_\infty \lesssim \|f\|_\infty^2.$$

Three steps to turn into existence proof:

(1) Solve linear perturbed problem:

$$\partial_t f = \nabla \cdot (A[g]\nabla f - \nabla a[g] f) + \delta \Delta f$$

for fixed $g \in L^1 \cap L^\infty$ using standard linear parabolic theory, which works since these conditions on g give bounded coefficients. Then compute estimate

$$\|f(t)\|_\infty \leq \|f_0\|_\infty \exp\left(\int_0^t \|g(s)\| \, ds\right).$$

We make rigorous the maximum principle by looking at

$$\frac{1}{2}\frac{d}{dt}\int (f-\ell)_+^2 \, dv = \int (f-\ell)_+ \partial_t f \, dv - \ell'(t) \int (f-\ell)_+ \, dv$$

Then use that f solves the equation and estimate

$$\frac{1}{2}\frac{d}{dt}\int (f-\ell)_+^2 \, dv \leq \frac{1}{2}\int (f-\ell)_+^2 g \, dv + \int [\ell g - \ell'(t)](f-\ell)_+ g \, dv$$

Then let $\ell(t)$ solve $\ell'(t) = \|g(t)\|_\infty \ell(t)$ and from the above can write

$$\frac{d}{dt}\int (f-\ell)_+^2\,dv \le \int (f-\ell)_+^2\,dv$$

Then if $\ell(0) \le \|f_0\|_\infty$, this starts zero, so by Grönwall is always zero and we can bound by

$$\ell(t) \le \|f_0\|_\infty \exp\left(\int_0^t \|g(s)\|\,ds\right).$$

(2) Show the solution operator has a fixed point in

$$K = \{f \in L^2(0,T;\mathbb{R}^3) : \|f\|_{L_t^\infty L_v^1} < C, 0 \le f \le C\}$$

with specified initial data. Here we need a short enough time that solution operator maps K to itself, this is determined by $\|f\|_\infty$.

(3) Take $\delta \to 0$ and use that our the estimates we did above are uniform in δ to get a solution to

$$\partial_t f = \nabla \cdot (A[f]\nabla f - \nabla a[f] f).$$

De Giorgi-Nash theory gives us that these solutions are smooth for positive time. Unique in class of weak solutions $L_t^1 L_v^\infty$ due to Fournier (2010) by probabilistic methods, alternate proof using Wasserstein by Caja-Lopez, Delgadino, Gualdani, and Taskovic.

4. Short Time Classical L^p Theory

Theorem. *(Golding-Loher 2023) If $f_0 \in L^p_m$ for $p > 3/2$, then the Landau equation instantly becomes bounded and smooth and exists up to a short time $T(\|f_0\|_{L^p_m})$.*

Idea is to use ODE argument to propagate L^p_m norm for a short time, test equation against $\langle v\rangle^k f^{p-1}$ to get the estimate

$$\frac{d}{dt}\langle v\rangle^k f^p\,dv + \frac{1}{2}\int \langle v\rangle^{k-3}|\nabla f^{p/2}|^2\,dv \lesssim \left(\int f^p\langle v\rangle^k\,dv\right)^{\frac{2p-1}{2p-3}}$$

Weighted norm deals with degeneracy of the ellipticity of A. Use energy structure to do a De Giorgi iteration to get pointwise control of f.

5. Landau-Bose-Einstein Extensions

We will consider the quantum Landau equation, where the quantum Landau operator is

$$\nabla \cdot \int \frac{\Pi(v-w)}{|v-w|}[f(w)(1+\theta\varepsilon f(w))\nabla_v f(v) - (1+\theta\epsilon f(v))f(v)\nabla_w f(w)]\,dw.$$

Here $\epsilon > 0$ and $\theta = \pm 1$. $\theta = -1$ is the fermionic case and the additional terms represent the Pauli exclusion principle, $\theta = 1$ is the bosonic case, the additional term represents the crowding effect of bosons. The quantum Landau operator was derived from quantum Boltzmann by Gualdani, Pavlovic, Toyota, and Wynter last year.
Fermi-Dirac case nicer than classical because we naturally can propagate a bound $f \le \varepsilon^{-1}$. Enables pre-Fisher proof of global smooth solutions in homogeneous case (Golding, Gualdani, Zamponi 2021) and more recent proof in inhomogeneous case

(Golding, Henderson 2026).
We consider bosonic case. Main result:

Theorem. *Let $f_0 \in L^\infty$ or L^p, $p > 3$. Then there exists a short time depending only on the norm for which LBE exists and smooths.*

Can normalize $\varepsilon = 1$ by rescaling f. Satisfies same conservation laws, but entropy is different:

$$S_B[f] := \int (1+f)\log(1+f) - f\log f \, dv$$

is monotone *increasing*. Worse, harder to understand functional. Unlike entropy, does not prevent concentration of mass into small sets, f can contain Dirac deltas.
Important because ellipticity estimate for $A[f]$ needs f to not concentrate too much. If f_n is an approximation to the identity, then $A[f_n] \to |v|^{-1}\Pi(v)$ which is not elliptic. So equation not necessarily uniformly elliptic in time.
For L^∞, write LBE in divergence form as

$$\partial_t f = \nabla \cdot (A[f(1+f)]\nabla f - f(1+f)\nabla a[f]).$$

Can use similar formal argument to say

$$\partial_t f_\infty \lesssim \|f\|_\infty^2 (1 + \|f\|_\infty).$$

More difficult to make rigorous due to additional nonlinearities, need specific choice of linearization. Argument generally goes through otherwise.
For L^p, more tricky because of ellipticity issue. Idea is to prove a quantitative lower bound for $A[f]$ in terms of $\|f\|_{L^\infty}$. Then do the same ODE propagation strategy, get a quantitatively worse time but not qualitatively worse result.